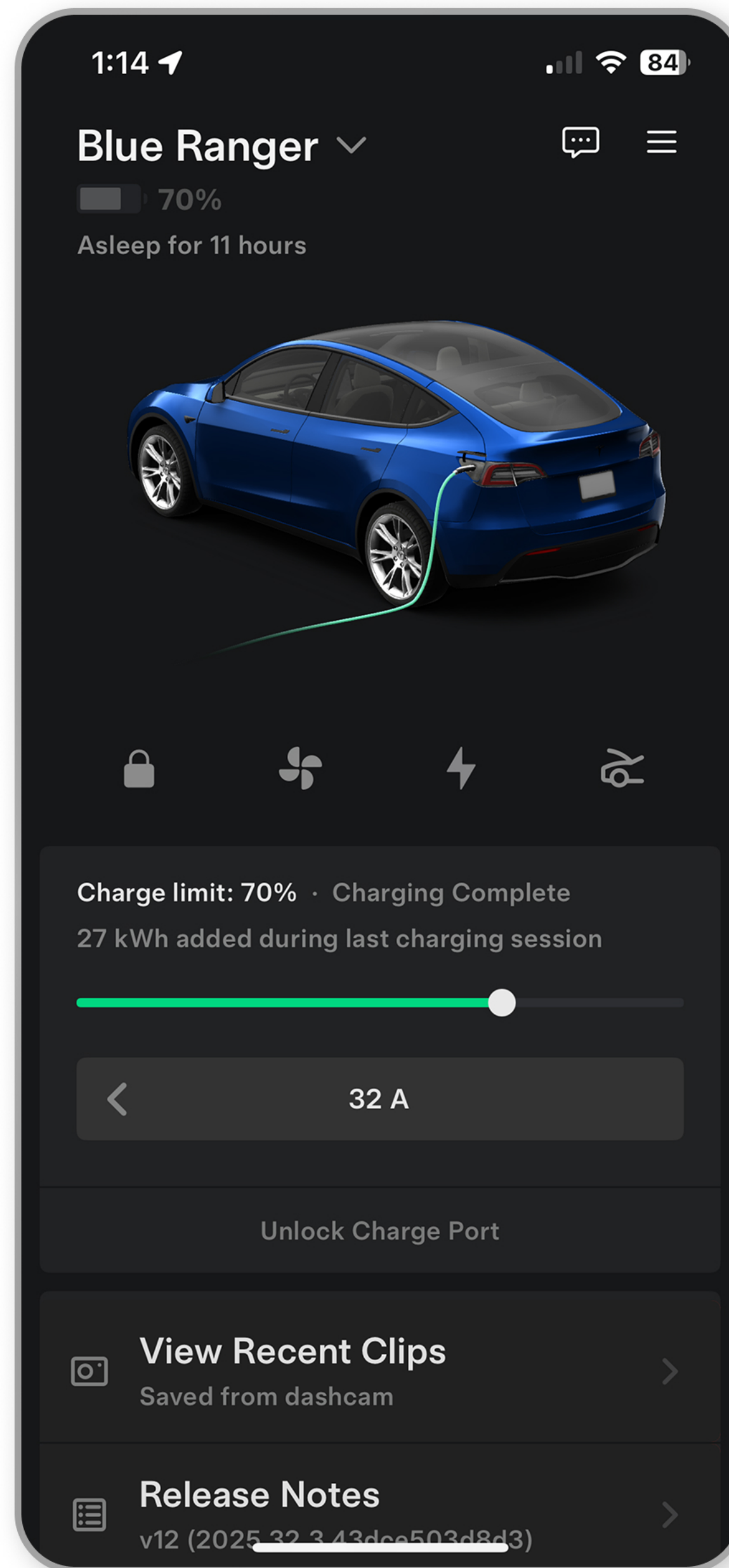


Amin Hassan



Tesla Off-Peak Charging Feature

Designing a smarter and more user friendly home charging experience



Introduction

Goal

Enable Tesla owners to charge effortlessly at the lowest cost and with **full confidence**, by automating off-peak charging and providing **clear savings insights** through the Tesla companion app

Reasoning

Right now, owners have to know their own off-peak hours and set them manually, which can be confusing or unreliable. The app does not show how much money or energy they are saving, so many people are unsure if scheduling is worth it. A smarter system would reduce hassle, build trust, and make charging feel effortless while supporting Tesla's mission of making EV adoption easier.

Vision

A Tesla app where home charging feels **automatic and stress-free**. The system adapts to utility rates, user preferences, and sustainability goals. Drivers just plug in, and Tesla takes care of the rest by saving money, reducing grid strain, and making the energy ecosystem smarter for everyone.

Current Problems

1

Lack of Support

The Tesla App current only lets users set charging times manually, but does not suggest cheapest off-peak times or show savings from off-peak charging

2

Reliance on User

To charge during off-peak times, the onus is on the user to find out their own off-peak hours, schedule it on the app, and calculate savings

3

Confusing UI

Users have reported frustrations with the scheduling UI, confusion between the start time and end time, and overall unreliability with the feature



User Testimonials

Tesla manual confirms there is a “finish by departure time” feature for reducing energy cost — but users still report ambiguity in savings and automation

“Why is scheduled charging as complicated as it is? if I don’t add a start time then charging starts immediately ... I have to do the backwards math ... and hope the schedule sticks until the next day.”

“My schedule keeps getting turned off, charging begins immediately upon plug in instead of my 2 AM setting.”

User Personas



Everyday Downtown Driver

25-35 years old. Works long hours with a long commute. Consistently charges his Tesla every day, but forgets to schedule it around off-peak hours to maximize savings. Seeks **automation**



Value Focused Owner

35-60 years old. Have lots of different trips a day with different route lengths. Demand a lot more from the battery and charge it more often. Seek to **save money** and **know how much they saved** to budget



Green User

25-45 years old. Bought a Tesla for their eco-friendliness first. Values the environment and reducing their carbon footprint. Seeks to charge their Tesla when the grid is the **cleanest** to reduce harm to the environment

All 3 of these personas share the following needs

Automation

Reliability

Transparency

Control

Key Takeaway

Focus on delivering all 4 shared needs in an easy-to-use UI

Current vs. Proposed Flow

To schedule charging during off-peak hours and maximize savings, here is what Tesla owners must do currently



Pain Points:

Manual Effort

No Savings Estimate

Occasional Faliure

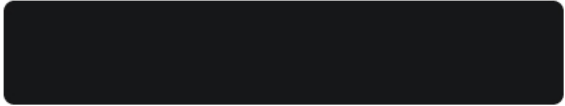
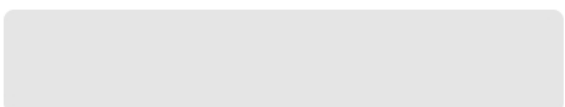
To address the pain points, here is a new proposed user flow to charge during off-peak hours



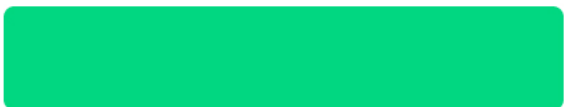
In this flow, the user only needs to confirm or override

Design System | Tokens

Surfaces

Background		#161718
Surface-Low		#212223
Surface-Medium		#333333
Icon-Primary		#E5E5E5
Icon-Secondary		#8A8B8B

Brand

Button-Primary		#3469FF
Button-Pressed		#204BC4
Status-Energy		#01D780

Text

Text-Primary		#FFFFFF
Text-Secondary		#848585
Text-Tertiary		#505151

Typography

Tesla Sans

Hero | Semi-Bold - 36pt

Drive The Future

Header | Semi-Bold - 24pt

Drive The Future

Body | Medium - 16pt

Drive The Future

Detail | Medium - 12pt

Drive The Future

Example

CA\$47.40

Example

Off-Peak Charging

Example

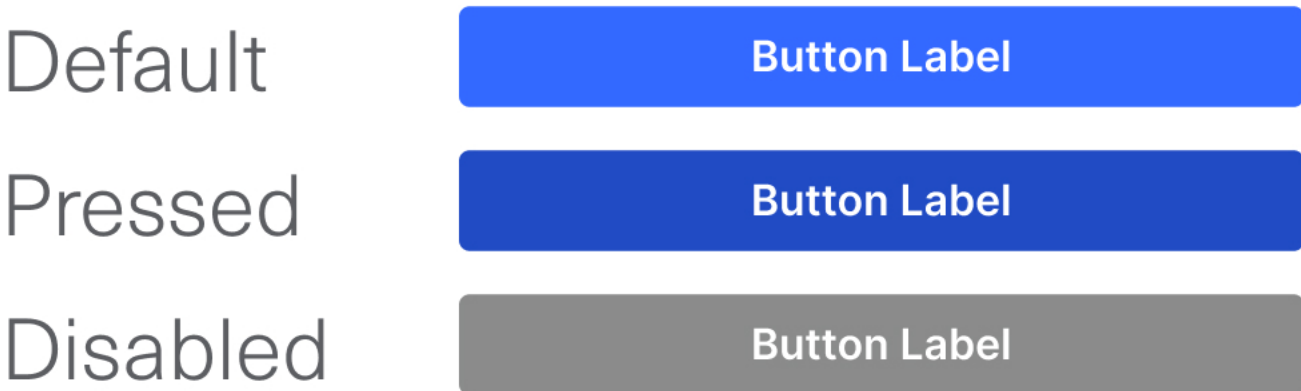
Utility Details

Example

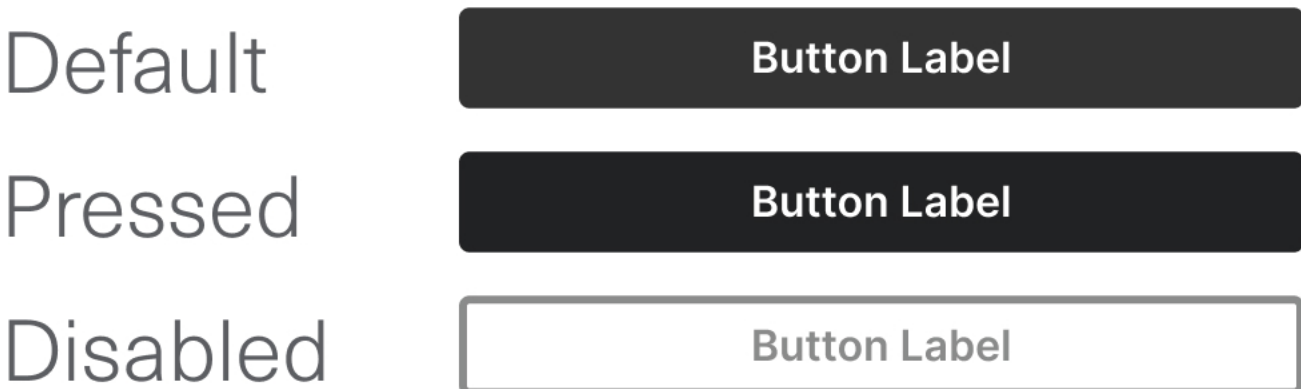
Off-Peak Hours

Design System | Components

Primary Button - Variants



Secondary Button - Variants

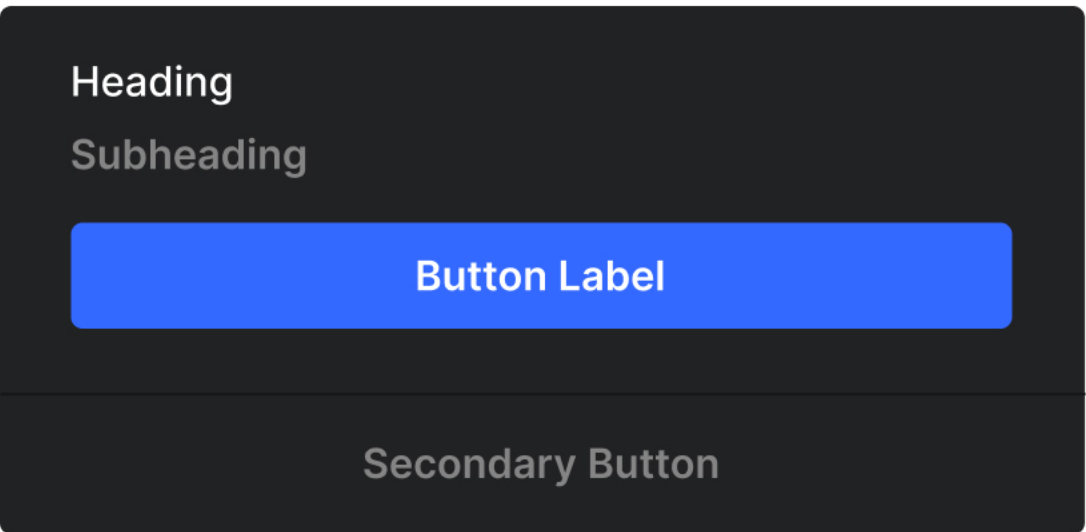


Toggle Switch - Variants

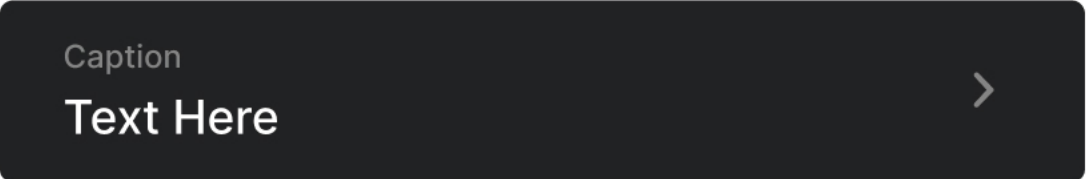


Standard Cards

Home Screen Card



Actionable Card



Segmented Control - Variants

Default



Pressed



Disabled



To address the user pain points, I propose 3 views, including one card prompt in the home screen and 2 detail screens

View 1

Home Screen

Initial Recommendation

View 2

Savings Screen

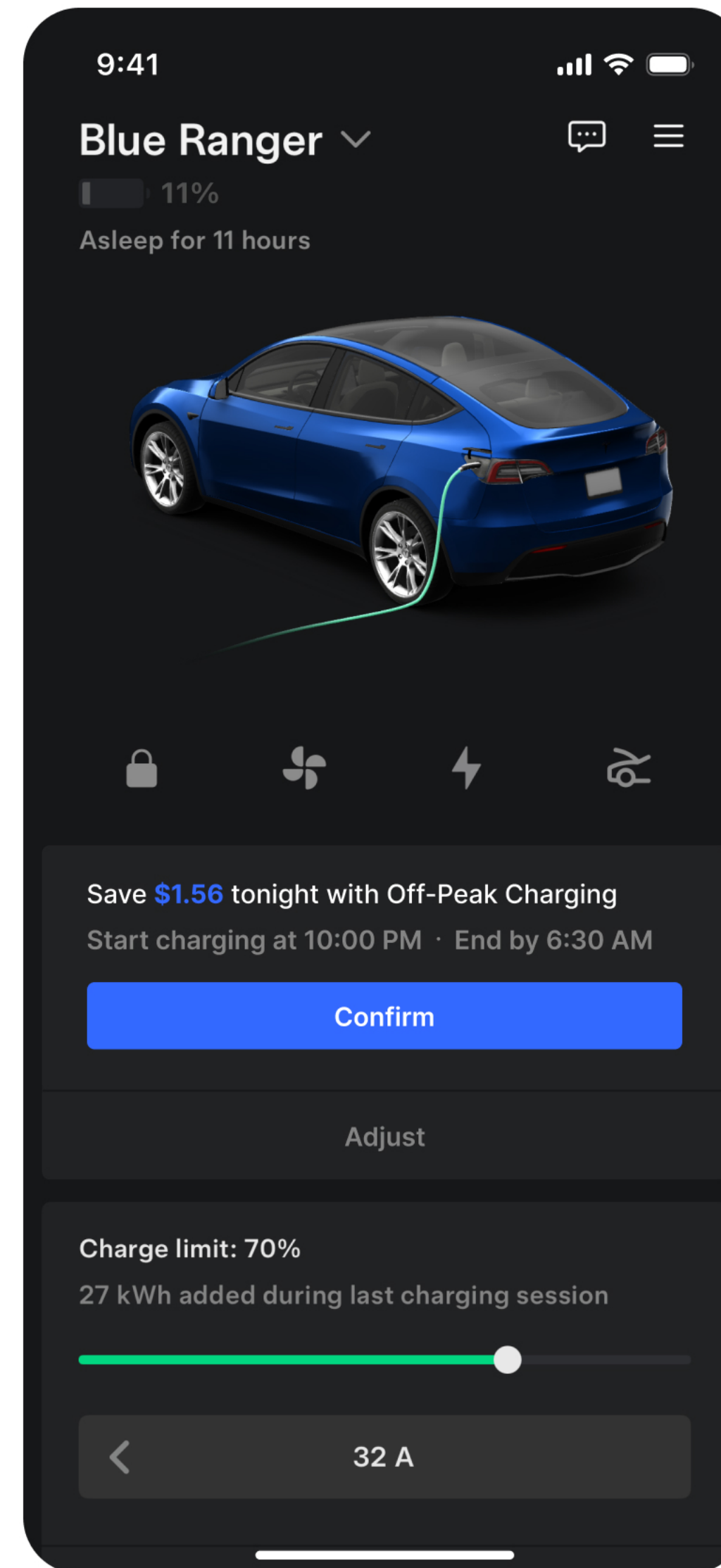
See how much you saved

View 3

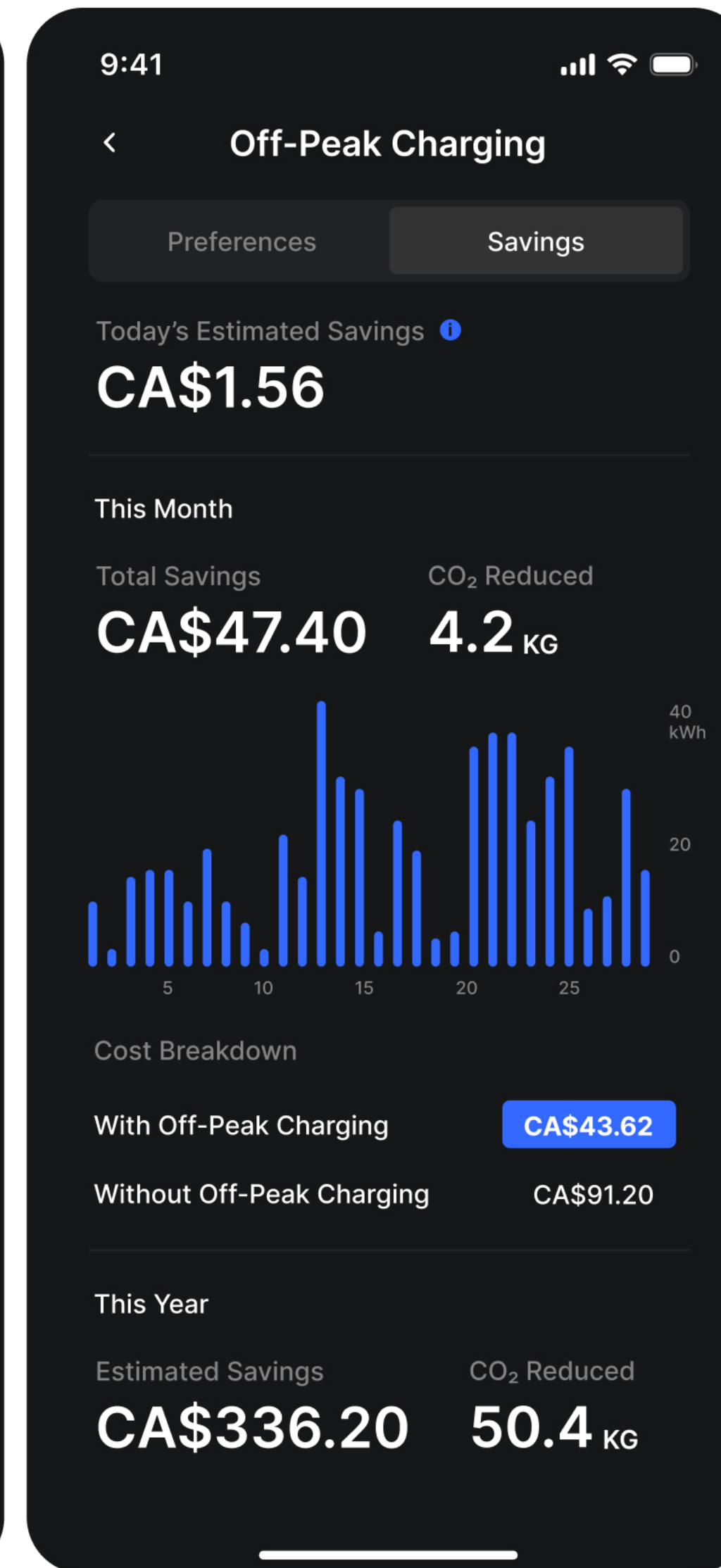
Preferences Screen

Adjust settings

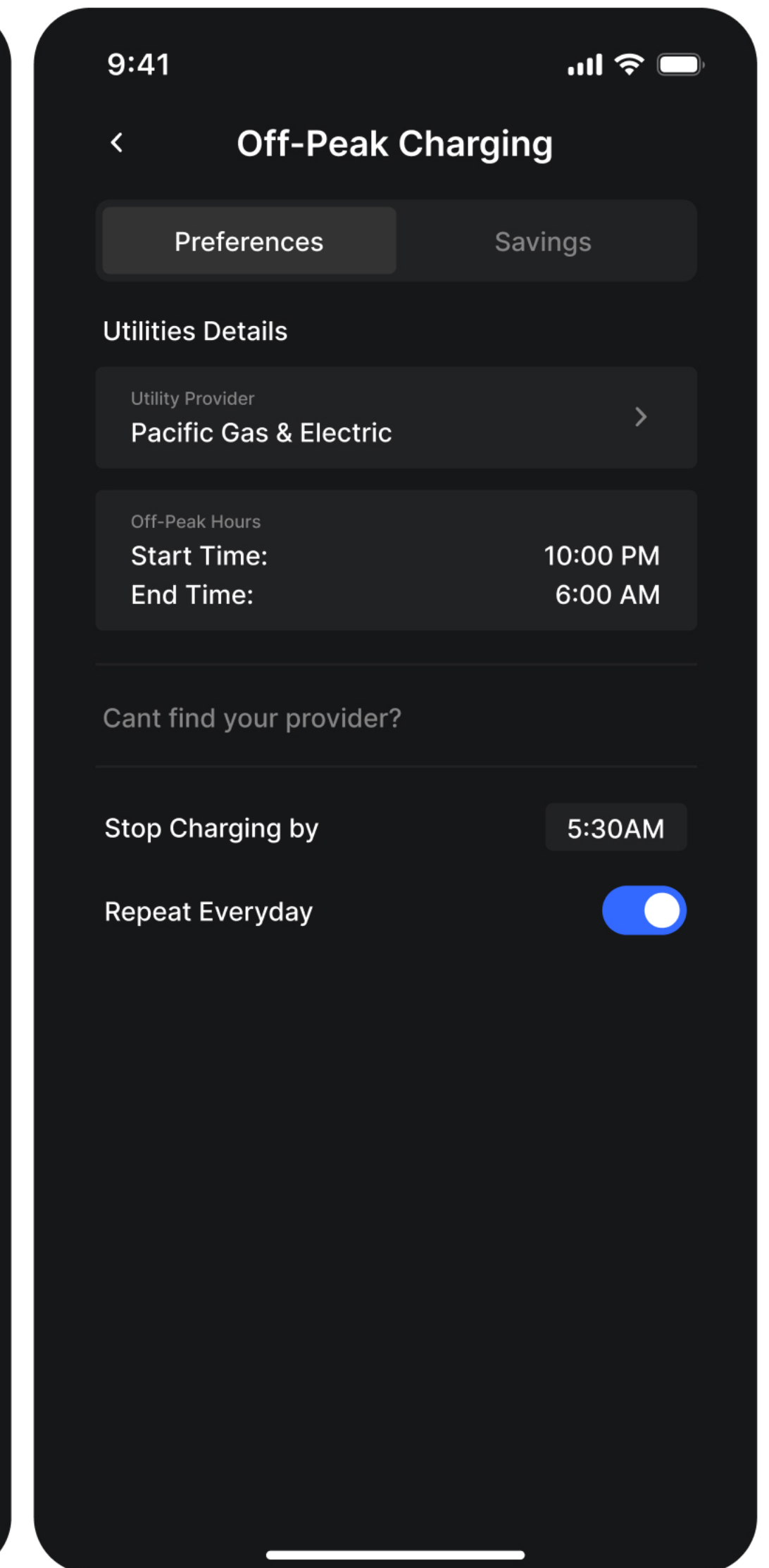
View 1



View 2



View 3



Impact & Next Steps

With the adoption of this feature, we should see the following benefits across these stakeholders

User Value

Saves Tesla drivers around **40%** on energy bills

Business Value

Increases Tesla brand loyalty and app stickiness

Social Value

Encourages off-peak charging, supporting grid sustainability

To ensure this feature continues meeting user needs and is validated in a proper manner

Prototype

Integrate utility rate APIs for real-time optimization and build interactive mockups

Measure

Track adoption, average savings, and reliability through app stats and user discourse

Iterate

Refine flows based on user feedback and edge cases (schedule reliability, savings clarity, etc...)

Amin Hassan | 3rd Year CS & BBA @ University of Waterloo

Thanks for viewing!

Find more at aminsubki.com